Reg. No. | | | | | | | | | | [17EC44]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

ANALOG ELECTRONIC CIRCUIT DESIGN-II

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. **Answer any FIVE full questions by choosing at least TWO questions from each part.**

ii. **Any missing data may be suitably assumed.**

PART - A

1. a. Draw the block diagram of op-amp and also explain its working. **08**

   b. What is meant by input offset current and offset voltage in an op-amp? **04**

   c. Find the time required for an input voltage change of 10 volts in a 741 op-amp. **02**

2. a. Draw the circuit diagram of difference amplifier and also derive the expression for output voltage. **07**

   b. Design a capacitor coupled inverting amplifier to have a voltage gain of 66. Given V_i=15mvolt and 741 op-amp is used. **07**

3. a. What is meant by frequency compensation and explain phase lag and phase lead compensation. **08**

   b. Calculate the slew rate limited cut off frequency for a voltage follower circuit using 741 op-amp if the peak of sinewave output is 5 volts. Also find the maximum peak value of output voltage if the voltage follower circuit is operating at 800 KHz unity gain cut off frequency. **06**

4. a. Draw the circuit diagram of Full wave precision rectifier and explain its working along with the necessary waveforms. **07**

   b. Design an inverting Schmitt trigger circuit to have an UTP = +3volts and LTP= -3 volts. Use 741 Op-amp with V_{CC}= 15 volts. **04**

   c. Draw the circuit diagram of an instrumentation amplifier and explain its working. **04**

PART - B

5. a. What is the need of stabilization circuit in an oscillator? Explain the working of stabilization circuit in an RC Phase shift oscillator. **08**

   b. Draw the circuit diagram of monostable multivibrator using IC 555 and explain its working along with the necessary waveforms. **06**

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6.	**a.**	What are the advantages of active filters?	**02**
	b.	Draw the block diagram of Band stop filter using band pass and summing circuit and also explain its working.	**05**
	c.	Using 741 op-amp, design a band pass filter to have a center frequency of 1KHz and a pass band of approximately ±33 HZ.	**07**
7.	**a.**	Draw the functional block diagram of LM723 voltage regulator and explain how it can be used as low voltage regulator.	**07**
	b.	What are the limitations of linear voltage regulator and how it can be eliminated in switching regulator?	**07**
8.	**a.**	Draw the diagram of inverted R-2R ladder network DAC and explain its working.	**07**
	b.	Draw the functional block diagram of successive approximation ADC and explain its working with an example.	**07**